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Task :

1 Movie List Manipulation program using c#

Solution :

using System;

namespace payodaHello

```
{  
    public enum enumtheater  
    {  
        vasu, vetri, msm  
    }  
}
```

public struct Movie

```
{  
    public int movie_id;  
    public string movie_name;  
    public int duration;  
    public enumtheater tname;  
}
```

public Movie(int movie_id, string movie_name, int duration, enumtheater tname)

```
{  
    this.tname = tname;  
    this.movie_id = movie_id;  
    this.movie_name = movie_name;  
    this.duration = duration;  
}
```

```
}
```

class Program

```
{  
    static void Main(string[] args)  
    {  
        Movie movie1 = new Movie(1, "RRR", 240, enumtheater.vasu);  
        Movie movie2 = new Movie(2, "Devara", 136, enumtheater.vetri);  
        Movie movie3 = new Movie(3, "Jurassic", 270, enumtheater.msm);  
        Movie movie4 = new Movie(4, "King", 140, enumtheater.vetri);  
        Movie movie5 = new Movie(5, "Devil", 220, enumtheater.vasu);  
    }  
}
```

```

List<Movie> list = new List<Movie>();
list.Add(movie1);
list.Add(movie2);
list.Add(movie3);
list.Add(movie4);
list.Add(movie5);

//foreach (Movie movie in list)
//{
//    Console.WriteLine( $"      Movie Id {movie.movie_id} Movie Name
{movie.movie_name} Duration Of Movie {movie.duration} Playing In {movie.tname} Theater");
//}

int total_movies = list.Count;
//Func<List<Movie>, int> totalMovies = list => list.Count;

Console.WriteLine("Total Movies: " + total_movies);
int totalDuration = list.Sum(x => x.duration);

//Func<List<Movie>, int> totalDuration = (List<Movie> movie) =>
//{
//    int sum = 0;
//    foreach (Movie movies in movie)
//    {
//        sum += movies.duration;
//    }
//    return sum;
//};

Console.WriteLine(" ");
Console.WriteLine("Total Duration: " + totalDuration);

int maxDuration =list.Max(m => m.duration);

Console.WriteLine(" ");
Console.WriteLine("Maximum Duration Movie : " + maxDuration + " minutes");

int minDuration =list.Min(m => m.duration);

Console.WriteLine(" ");
Console.WriteLine("Minimum Duration Movie : " + minDuration + " minutes");

```

```

Movie movie = list.OrderBy(m => m.duration).First();

Console.WriteLine(" ");
Console.WriteLine( $"Shortest Movie => Movie Id {movie.movie_id} Movie Name
{movie.movie_name} Duration Of Movie {movie.duration} Playing In {movie.tname} Theater");

Console.WriteLine(" ");

Movie longmovie = list.OrderByDescending(m => m.duration).First();

Console.WriteLine( $"Longest Movie => Movie Id {longmovie.movie_id} Movie Name
{longmovie.movie_name} Duration Of Movie {longmovie.duration} Playing In {longmovie.tname}
Theater");

Console.WriteLine(" ");
var vetriMovies = list.Where(m => m.tname == enumtheater.vetri);
Console.WriteLine("Vetri Theater Movie: ");
Console.WriteLine(" ");
foreach (var m in vetriMovies)
{
    Console.WriteLine( $"    {m.movie_name}, {m.duration} min");
}

Console.WriteLine(" ");
var longMovies = list.Where(m => m.duration > 150);
Console.WriteLine("Long Movie : ");
Console.WriteLine(" ");
foreach (var m in longMovies)
{
    Console.WriteLine( $"    {m.movie_name}, {m.duration} min");
}

Console.WriteLine(" ");
Console.WriteLine("Movie List From The Lower To Higher Duration :");
Console.WriteLine(" ");
var shorttolong = list.OrderBy(m => m.duration);
foreach (var m in shorttolong)
{
    Console.WriteLine( $"    Movie Name: {m.movie_name}, Duration: {m.duration} min");
}

Console.WriteLine(" ");
Console.WriteLine("Top Movie Name List :");

```

```

        Console.WriteLine(" ");
        var top3 = list.OrderByDescending(m => m.duration).Take(3);
        foreach (var m in top3)
            Console.WriteLine($"    {m.movie_name} - {m.duration} min");
    }
}
}

```

Output:

```

Microsoft Visual Studio Debug Console
Total Movies: 5
Total Duration: 1006
Maximum Duration Movie : 270 minutes
Minimum Duration Movie : 136 minutes
Shortest Movie => Movie Id 2 Movie Name Devara Duration Of Movie 136 Playing In vetri Theater
Longest Movie => Movie Id 3 Movie Name Jurassic Duration Of Movie 270 Playing In msm Theater
Vetri Theater Movie:
    Devara, 136 min
    King, 140 min
Long Movie :
    RRR, 240 min
    Jurassic, 270 min
    Devil, 220 min
Movie List From The Lower To Higher Duration :
    Movie Name: Devara, Duration: 136 min
    Movie Name: King, Duration: 140 min
    Movie Name: Devil, Duration: 220 min
    Movie Name: RRR, Duration: 240 min
    Movie Name: Jurassic, Duration: 270 min
Top Movie Name List :
    Jurassic - 270 min
    RRR - 240 min
    Devil - 220 min
C:\Users\narendrakumar.s\source\repos\ConsoleApp3\ConsoleApp3\bin\Debug\net8.0\ConsoleApp3.exe (process 14772) exited with code 0 (0x0).
Press any key to close this window . . .

```

